



RAG ARCHITECTURE

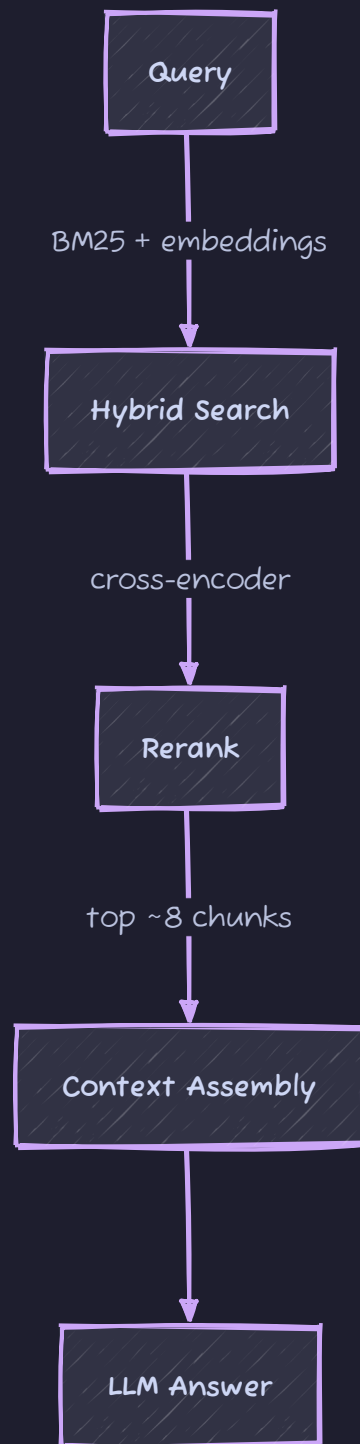
Your RAG pipeline doesn't have a model problem.

It has a retrieval problem — hybrid search + reranking took answer accuracy 2.3x higher, same LLM, same index.





What A Retrieval Layer That Actually Works Looks Like



Why Naive Top-K Retrieval Breaks First



Pure vector search misses exact keyword matches — IDs, error codes, product SKUs — that BM25 catches trivially.



Fixed-size chunking slices sentences mid-thought, so the "most similar" chunk is missing the clause that actually answers the question.



No reranking step means the top-k the LLM sees is ranked by embedding similarity alone, not the query's real intent.



Recall gets "fixed" by raising k instead of raising precision — the model drowns in more noisy context, not better context.



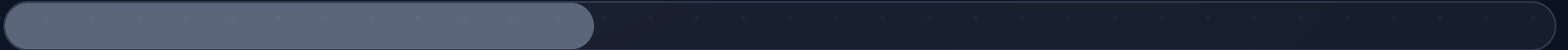


2.3x

Higher answer accuracy

Same base vector index, same LLM, same chunk size — only the retrieval layer changed: added BM25+embedding hybrid search and a cross-encoder rerank stage on top.

Vector-only top-k – recall@10



Hybrid + rerank – recall@10



Fix It In This Order

- 01 Add BM25 (or another sparse/keyword method) alongside your vector index — hybrid retrieval, not vector-only.

- 02 Rerank the top ~50 hybrid candidates with a cross-encoder before the LLM ever sees them.

- 03 Chunk semantically — headings, paragraphs, code blocks — instead of a fixed character count.

- 04 Only widen the context window after retrieval precision is fixed. A bigger window just means more room for the wrong chunks.





Fix retrieval before you touch the model.

Hybrid search → rerank → semantic chunking → then widen the context window. Follow for the next breakdown in this series.

